

Project 02

AI Music & Audio Analyzer

Living Project Documentation — v0.3

Item	Current Value
Project Status	In Progress — Audio Capture, Time/Frequency Analysis Verified
Platform	Windows Laptop
Python	3.14.7
Environment	VS Code + PowerShell + Python venv
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Audio Input	Airdopes Joy Hands-Free microphone
Audio Output	Airdopes Joy Stereo
Analysis Libraries	sounddevice + Librosa + NumPy + Matplotlib
Recorded Test Signal	5-second mono WAV

1. Project Definition

AI Music & Audio Analyzer is a Python-based project for real hardware audio capture, digital signal processing, visualization, frequency analysis, music/audio feature extraction, and later AI interpretation. The Windows laptop is the processing platform and Airdopes Joy Bluetooth earbuds provide the microphone input and playback path.

2. Current Architecture

Airdopes Joy Microphone → Bluetooth → Windows Audio System → Python sounddevice → WAV Capture → Librosa / NumPy → Signal Analysis → Matplotlib Visualizations → Reports / Future Dashboard

3. Completed Milestones

Step	Milestone	Status
1	Audio device detection	PASS
2	Airdopes microphone verification	PASS
3	Five-second microphone recording	PASS
4	WAV loading and basic analysis	PASS
5	Waveform visualization	PASS
6	Audio level analysis	PASS
7	FFT / frequency spectrum analysis	PASS
8	STFT / spectrogram analysis	PASS

4. Step 1 — Audio Device Detection

Windows audio-device enumeration successfully detected the Airdopes Joy Bluetooth endpoints. Device 2 was identified as the Airdopes Joy Hands-Free microphone with one input channel and a reported 44.1 kHz rate. Device 4 was identified as Airdopes Joy Stereo with two output channels at 44.1 kHz.

5. Step 2 — Airdopes Microphone Verification

The Airdopes Joy Hands-Free microphone was selected through Python sounddevice and produced a non-zero signal during the microphone test, confirming that the Bluetooth microphone is accessible to the Python application.

6. Step 3 — Airdopes Audio Recording

A five-second mono recording was successfully captured from device 2 and saved as audio\airdopes_test.wav.

Parameter	Result
Device ID	2
Input Channels	1
Sample Rate	44,100 Hz
Duration	5.00 seconds
Samples	220,500
RMS Amplitude	0.031082
Peak Amplitude	0.423889

7. Step 4 — WAV Audio Analysis

The saved WAV file was loaded successfully with Librosa. The measurements matched the recording stage, confirming reliable file processing.

Sample Rate: 44,100 Hz | Channels: 1 | Samples: 220,500 | Duration: 5.00 sec

RMS: 0.031082 | Peak: 0.423889 | RMS: -30.15 dBFS | Peak: -7.45 dBFS

8. Step 5 — Waveform Visualization

The recorded signal was converted into a time-domain waveform with Librosa and Matplotlib. The graph was successfully generated and saved as reports\airdopes_waveform.png.

9. Step 6 — Audio Level Analysis

The audio-level analyzer calculated amplitude statistics, RMS, peak, dBFS levels, crest factor, and RMS level over time.

Measurement	Result
Minimum Amplitude	-0.347687
Maximum Amplitude	0.423889
Mean Absolute Amplitude	0.010442
RMS Amplitude	0.031082
Peak Amplitude	0.423889
RMS Level	-30.15 dBFS
Peak Level	-7.45 dBFS
Crest Factor	13.638
Level Classification	Healthy
Graph	reports\airdopes_audio_level.png

10. Step 7 — FFT / Frequency Spectrum Analysis

A Fast Fourier Transform was applied to the five-second Airdopes recording. A Hann window was applied before the real-valued FFT. The frequency spectrum was calculated and visualized.

FFT Measurement	Actual Result
Sample Rate	44,100 Hz
Samples	220,500
FFT Output Samples	110,251
Frequency Resolution	0.2000 Hz
Maximum Frequency / Nyquist	22,050 Hz
Dominant Frequency	544.60 Hz
Dominant Magnitude	0.001539
Graph	reports\airdopes_frequency_spectrum.png

The strongest detected FFT bin was approximately 544.60 Hz. This result represents the strongest frequency component in this particular microphone test recording; it is not by itself a musical pitch classification.

11. Step 8 — STFT / Spectrogram Analysis

Short-Time Fourier Transform (STFT) was applied to observe how frequency content changes over time. The spectrogram uses an FFT size of 2048 and a hop length of 512.

STFT Measurement	Actual Result
Sample Rate	44,100 Hz
FFT Size	2048
Hop Length	512
Frequency Bins	1025
Time Frames	431
Frequency Resolution	21.53 Hz
Maximum Frequency	22,050 Hz
Graph	reports\airdopes_spectrogram.png

The spectrogram provides a time-frequency representation of the captured audio. The displayed visualization is limited to 0–10 kHz for easier inspection, while the underlying STFT reaches the 22.05 kHz Nyquist frequency.

12. Current Analysis Outputs

Output	Purpose	Status
audio\airdopes_test.wav	Raw captured microphone recording	Generated
reports\airdopes_waveform.png	Amplitude versus time	Generated
reports\airdopes_audio_level.png	RMS level versus time	Generated
reports\airdopes_frequency_spectrum.png	Frequency versus magnitude	Generated
reports\airdopes_spectrogram.png	Frequency versus time	Generated

13. Current Project Structure

Project-02-AI-Music-Audio-Analyzer/

■■■ audio/ → airdopes_test.wav

■■■ reports/ → waveform, audio-level, frequency-spectrum and spectrogram images

■■■ src/ → audio_devices.py, test_airdopes_mic.py, record_airdopes.py, analyze_airdopes_audio.py, plot_airdopes_waveform.py, audio_level_analysis.py, fft_frequency_analysis.py, spectrogram_analysis.py

■■■ tests/

■■■ venv/

■■■ requirements.txt

■■■ README.md

14. Testing Record

Test	Result
Audio device enumeration	PASS
Airdopes Joy microphone detection	PASS
Airdopes Joy stereo output detection	PASS
Microphone signal capture	PASS
Five-second WAV generation	PASS
WAV loading with Librosa	PASS
Waveform generation	PASS
Audio level analysis	PASS
RMS level graph generation	PASS
FFT / frequency spectrum	PASS
Dominant frequency extraction	PASS — 544.60 Hz
STFT / spectrogram	PASS

15. Next Development Step

Step 9 — Spectral Feature Analysis. The next module will extract music/audio-oriented spectral features such as spectral centroid, spectral bandwidth, spectral rolloff, spectral flatness, and zero-crossing rate. These features will prepare the project for bass/mid/treble analysis and later music classification.

16. Version History

v0.1 — 19-Aug-2026: New documentation baseline created from the beginning. Hardware setup, Python environment, audio-device detection, Airdopes microphone verification and recording documented.

v0.2 — 19-Aug-2026: Steps 4–6 completed. WAV loading, waveform visualization, RMS/peak/dBFS analysis, amplitude statistics, crest factor and RMS level graph documented with actual measurements.

v0.3 — 19-Aug-2026: Steps 7–8 completed. FFT frequency spectrum and STFT spectrogram analysis documented with actual FFT resolution, dominant frequency, STFT configuration, frequency bins and time frames. Generated frequency-spectrum and spectrogram outputs recorded.

Only the new Project-02 history is used; previous report versions are intentionally excluded.